Collected Mathematical Papers, Vol. XII

PDF pages 301–325

Papers 831 (tables, concl.), 832, 833, 834, 835, 836, 837 (start)

Arthur Cayley

831 (continued). Seminvariant Tables.

[The pages 281–289 of the printed volume (PDF pp. 301–309) consist almost entirely of the numerical seminvariant tables which form the substance of paper 831. The tables are very large and densely populated with small integer coefficients; their precise transcription as ASCII tabulars is impractical without risk of arithmetical corruption. The structure is summarized below; the original scan should be consulted for the exact entries.]

281 **SEMINVARIANT TABLES.** **[831**

Continuation of the principal table for ($k = 10$)*, with rows headed* k, bj, ci, dh, eg, f^2, b^2i, bch, bdg, bef, cdf, ce^2, d^2e, $bcde$, bd^3, c^3e, c^2d^2, b^3de, b^3d^2, bc^3d, c^5, b^2c^3, b^4cd, b^6, *etc.; and columns headed* k, ci, dh, eg, f^2, c^2f, cde, cd^3, c^3e, c^2d^2, c^5.

[Table for $k = 10$*: dense triangular array of small signed integers (typical magnitudes* 1–210*). Footer carries the printer's signature "C. XII." and gathering number "36".]*

Table for ($l = 11$), with row labels l, bk, cj, di, eh, fg, b^2j, bci, bdh, beg, bf^2, c^2h, cdg, cef, d^2f, de^2, b^3i, …, $bcdf$, bce^2, bd^2e, c^3f, c^2de, cd^3 and column labels l, cj, di, eh, fg, c^2h, cdg, cef, d^2f, c^3f, de^2, c^2de, cd^3, c^2d.

[Table for $l = 11$: dense triangular array of small signed integers (range ≈ -420 to $+810$). At the foot: "(Continued next page.)".]

(Continuation of the $l = 11$ table.) Row labels include b^3cg, b^2fg, b^2cdg, b^2cef, b^2d^2f, b^2de^2, bc^3f, bc^2de, bcd^3, c^4e, c^3d^2, b^4f, b^4cg, b^4de, b^4d^2, b^3c^2f, b^3cde, b^3d^3, b^2c^4, b^6e, b^6cd, b^8, etc.

[Continuation of the $l = 11$ table: lower portion of the same dense triangular array. Footer "36–2".]

Table for ($m = 12$). The leading row labels are m, bl, ck, dj, ei, fh, g^2, b^2k, bcj, bdi, beh, bfg, c^2i, cdh, ceg, cf^2, d^2g, def, e^3, b^3j, b^2ci, b^2dh, b^2eg, b^2f^2, bc^2h, $bcdg$, $bcef$, bde^2, c^3g, …, cd^2e, d^4, b^4i, b^3ch, b^3dg, b^3ef, …; and column labels m, ck, dj, ei, fh, g^2, c^2i, cdh, ceg, d^2g, cf^2, def, e^3, c^3g, c^2df, cd^2e, d^4, c^4e, c^3d^2, …

[Table for $m = 12$, top portion: very large dense triangular array; entries are small signed integers typically in the range -792 to $+925$, including factors such as ± 462, ± 400, ± 350. Foot: "(Continued next page.)".]

(Continuation of the $m = 12$ table.) Row labels continue with b^2cdf, b^2ce^2, b^2d^2e, b^2d^3, bc^3f, bc^2de, bcd^3, c^4f, c^3de, c^2d^3, b^4h, b^3cg, b^3df, b^3e^2, b^2c^2g, b^2cdf, b^2ce^2, b^2d^2e, b^2d^3, bc^4f, bc^3de, bc^2d^3, c^6, b^6g, ...

[Continuation of the $m = 12$ table: lower portion of the same large triangular array. Sparse entries continue, with characteristic large coefficients (e.g., ± 332, ± 199, ± 144).]

Subsidiary Tables: $b = 0$.

Left-hand: up to ($m = 12$).

Col. c

c	$+1$

Col. d

d	$+1$

Col. e, c^2

e	$+1$	$+3$
c^2		$+1$

Col. f, cd

f	$+1$	$+9$
cd		$+1$

Col. g, ce, d^2, c^3

g	$+1$	$+15$	-10	
ce		-1	-1	-1
d^2			$+1$	$+4$
c^3				$+1$

Col. h, cf, de, c^2d

h	$+1$	$+9$	-8	
cf		-1	-1	-1
de			$+1$	$+5$
c^2d				$+1$

[Further subsidiary tables for columns labelled by other monomials of weight up to 12, with entries typically in $\{-60, \ldots, +60\}$. The detailed cell-by-cell transcription is omitted; entries are most safely consulted from the scan.]

Right-hand: up to $(k = 10)$.

$$\frac{+2}{c}\ \left|\ \begin{array}{ll} c & -2 \end{array}\right. \qquad \frac{+6}{d}\ \left|\ \begin{array}{ll} d & -3 \end{array}\right. \qquad \frac{+24}{e\ c^2}\ \left|\ \begin{array}{lrr} e & -4 & +12 \\ c^2 & +2 & +\ \ 6 \end{array}\right. \qquad \frac{+120}{f\ cd}\ \left|\ \begin{array}{lrr} f & -5 & +60 \\ cd & +3 & +10 \end{array}\right.$$

$$\frac{+720}{g\ ce\ d^2\ c^3}\ \left|\ \begin{array}{lrrrr} g & -6 & +90 & -60 & -180 \\ ce & +6 & -30 & -60 & -180 \\ d^2 & +3 & -15 & +60 & +\ \ 90 \\ c^3 & -2 & -10 & +20 & \end{array}\right.$$

[Further right-hand subsidiary tables, with row of denominator $+5040$ *for* h, cf, de, c^2d*;* $+40320$ *for* i*;* $+362880$ *for* j*; and* -3628800 *for* k*; precise entries omitted as above.]*

Left-hand: continuation. Cols. i, cg, dh, ef, f^2, c^2g, cdf, ce^2, d^2e, c^3e, c^2d^2, c^5.

[Triangular table with row labels $i, cg, dh, ef, f^2, c^2g, cdf, ce^2, d^2e, c^3e, c^2d^2, c^5$ and identical column labels. Entries include +1, +45, −120, +210, −126, and smaller signed integers down the lower triangle.]

Right-hand: denominator +3628800.

[Right-hand triangular table with the same row/column labels and the indicated denominator. Entries are large signed integers (e.g., −10, +450, +1230, −2100, −22000, etc.).]

Left-hand and right-hand continuations. Cols. l, cj, di, eh, fg, c^2h, cdg, cef, d^2f, de^2, c^3f, c^2de, cd^3, c^2d *and* m, ck, dj, ei, fh, g^2, c^2i, cdh, ceg, cf^2, d^2g, def, e^3, c^3g, c^2df, cd^2e, d^4, c^4e, c^3d^2.

[Two large dense triangular tables (left-hand for weight 11*, right-hand for weight* 12*). Entries are signed integers, several rows headed "no right-hand" indicating that the corresponding row receives no contribution to the right-hand member. Footer: "C. XII." and gathering "37".]*

832.

NOTE ON AN APPARENT DIFFICULTY IN THE THEORY OF CURVES, WHEN THE COORDINATES OF A POINT ARE GIVEN AS FUNCTIONS OF A VARIABLE PARAMETER.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 12–14.]

SUPPOSE that the homogeneous coordinates x, y, z are given as proportional to the following functions of a parameter λ,

$$x : y : z = u + \alpha\sqrt{\Omega},\ \ v + \beta\sqrt{\Omega},\ \ w + \gamma\sqrt{\Omega},$$

where u, v, w are linear functions, Ω a cubic function, of the parameter. For the intersections of the curve with the arbitrary line $Ax + By + Cz = 0$, we have

$$Au + Bv + Cw + (A\alpha + B\beta + C\gamma)\sqrt{\Omega} = 0,$$

that is,

$$(Au + Bv + Cw)^2 - (A\alpha + B\beta + C\gamma)^2\Omega = 0,$$

a cubic equation in λ; and the curve is thus a cubic. For the value $\lambda = \infty$ we have $x : y : z = \alpha : \beta : \gamma$, or the point (α, β, γ) is a point of the curve.

Suppose now that the line $Ax + By + Cz = 0$ is an arbitrary line through the point (α, β, γ); viz. let the coefficients A, B, C satisfy the relation $A\alpha + B\beta + C\gamma = 0$: the equation for the determination of λ becomes

$$(Au + Bv + Cw)^2 = 0,$$

which equation has two equal roots, suppose $\lambda = \lambda_0$; and the meaning of this is not at once obvious.

Observe that more properly there is a root $\lambda = \infty$ which has dropped out, and that the roots are $\lambda = \infty$, $\lambda = \lambda_0$, $\lambda = \lambda_0$. The root $\lambda = \infty$ gives the point (α, β, γ), which is of course one of the intersections of the line with the curve. The two roots λ_0 give not the same intersection but two different intersections of the line with the curve; the line being in fact a line through the point (α, β, γ) of the curve, and which besides meets the curve in two distinct points.

To see how this is, observe that, in the general case where $A\alpha+B\beta+C\gamma$ is not $=0$, we have λ determined by a cubic equation as above; and then taking λ equal to any root of this equation, we have further

$$Au+Bv+Cw+(A\alpha+B\beta+C\gamma)\sqrt{\Omega}=0,$$

viz. the value of $\sqrt{\Omega}$ is hereby uniquely determined; and to each of the three values of λ, $\sqrt{\Omega}$, there corresponds a determinate point (x,y,z).

But suppose now $A\alpha+B\beta+C\gamma=0$, and λ determined by the equation

$$(Au+Bv+Cw)^2=0,$$

giving $\lambda=\lambda_0$ as above. There is no longer an equation for the unique determination of $\sqrt{\Omega}$, and to the value $\lambda=\lambda_0$ there correspond the two values $\sqrt{\Omega_0}$, $-\sqrt{\Omega_0}$ of the radical: and thus to the two roots $\lambda=\lambda_0$, $\lambda=\lambda_0$ correspond the two different points

$$x:y:z=u_0+\alpha\sqrt{\Omega_0}:v_0+\beta\sqrt{\Omega_0}:w_0+\gamma\sqrt{\Omega_0}$$

and

$$x:y:z=u_0-\alpha\sqrt{\Omega_0}:v_0-\beta\sqrt{\Omega_0}:w_0-\gamma\sqrt{\Omega_0}.$$

It is to be added that the point (α,β,γ) is an inflexion on the curve. Write for a moment

$$u,v,w=a\lambda+f,\ b\lambda+g,\ c\lambda+h,$$

and let A, B, C be determined by the conditions

$$A\alpha+B\beta+C\gamma=0,\qquad Aa+Bb+Cc=0.$$

Then the equation for the determination of λ becomes $(Af+Bg+Ch)^2=0$, viz. the left-hand is a mere constant, or there are the three equal roots $\lambda=\infty$; the intersections with the curve are thus the point (α,β,γ) three times; hence this point is an inflexion, the tangent being $Ax+By+Cz=0$. The second of the two equations may be written

$$Au+Bv+Cw=0.$$

Let λ_1 be one of the roots of the equation $\Omega=0$; u_1, v_1, w_1 the corresponding values of u, v, w, and let A_1, B_1, C_1 be determined by the conditions

$$A_1\alpha+B_1\beta+C_1\gamma=0,\qquad A_1u_1+B_1v_1+C_1w_1=0.$$

The equation $(Au+Bv+Cw)^2=0$ for the intersections with the curve has the two equal roots $\lambda=\lambda_1$; and to each of these, since now $\sqrt{\Omega(\lambda_1)}=0$, there corresponds the same point $x:y:z=u_1:v_1:w_1$; hence the line $A_1x+B_1y+C_1z=0$, or say

$$A_1x+B_1y+C_1z=0,$$

is a tangent from the inflexion. Similarly, if λ_2, λ_3 are the other two roots of the equation $\Omega=0$, we have $A_2x+B_2y+C_2z=0$, $A_3x+B_3y+C_3z=0$ for the other two tangents from the inflexion.

It would have been to some extent clearer to have represented the parameter λ as a quotient, say $\lambda=p/q$; the equations for x, y, z would then have been

$$x:y:z=(ap+fq)\sqrt{q}+\alpha\sqrt{\Omega}:(bp+gq)\sqrt{q}+\beta\sqrt{\Omega}:(cp+hq)\sqrt{q}+\gamma\sqrt{\Omega},$$

where Ω is now a homogeneous function $(p,q)^3$.

833.

ON A FORMULA IN ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 21, 22.]

WRITING s, c, d for the sn, cn, and dn of an argument u, and so in other cases: we have s, c, d for the coordinates of a point on the quadriquadric curve $x^2+y^2=1$, $z^2+k^2x^2=1$. Applying Abel's theorem to this curve, it appears that, if $u_1+u_2+u_3+u_4=0$, the corresponding points are in a plane; that is, the elliptic functions satisfy the relation

$$\begin{vmatrix} s_1, & c_1, & d_1, & 1 \\ s_2, & c_2, & d_2, & 1 \\ s_3, & c_3, & d_3, & 1 \\ s_4, & c_4, & d_4, & 1 \end{vmatrix} = 0.$$

This may be written

$$\begin{aligned}(s_1-s_4)(c_2d_3-c_3d_2)+(s_2-s_3)(c_4d_1-c_1d_4)\\ +(c_1-c_4)(d_2s_3-d_3s_2)+(c_2-c_3)(d_4s_1-d_1s_4)\\ +(d_1-d_4)(s_2c_3-s_3c_2)+(d_2-d_3)(s_4c_1-s_1c_4)=0;\end{aligned}$$

and it may be shown that each of the three lines is, in fact, separately $=0$.

This appears from the following three formulæ:

$$\frac{\operatorname{sn}(u_1+u_2)}{\operatorname{cn}(u_1+u_2)-\operatorname{dn}(u_1+u_2)}=\frac{s_1-s_2}{c_1d_2-c_2d_1},$$

$$\frac{\operatorname{sn}(u_1+u_2)}{\operatorname{cn}(u_1+u_2)+1}=\frac{c_1-c_2}{d_1s_2-d_2s_1},$$

$$\frac{\operatorname{sn}(u_1+u_2)}{\operatorname{dn}(u_1+u_2)+1}=-\frac{1}{k^2}\,\frac{d_1-d_2}{s_1c_2-s_2c_1},$$

which are themselves at once deducible from formulæ given, p. 63, of my *Elliptic Functions*, and which may be written

$$\begin{aligned}\operatorname{sn}(u_1+u_2)\big(s_1^2s_2^2-c_2^2\big)&=-(s_1^2-s_2^2)(c_1c_2-s_1s_2d_1d_2)\ \div\ (c_1d_2-c_2d_1),\\ \operatorname{cn}(u_1+u_2)&=s_1c_1d_2\ \div\ \ldots,\\ \operatorname{dn}(u_1+u_2)&=s_2c_2d_1\ \div\ \ldots.\end{aligned}$$

[The right-hand sides of the last two formulæ involve the same denominator and similar numerator structures; the precise typography is partly obscured in the scan.]

In fact, the numerators of $\mathrm{cn}(u_1+u_2)-\mathrm{dn}(u_1+u_2)$, $\mathrm{cn}(u_1+u_2)+1$, $\mathrm{dn}(u_1+u_2)+1$ thus become $=(s_1-s_2)(d_1d_2-c_1c_2)$, $(c_1-c_2)(d_1s_2-d_2s_1)$, $(d_1-d_2)(s_1c_2-s_2c_1)$ respectively: so that, taking the numerator of $\mathrm{sn}(u_1+u_2)$ successively under its three forms, we have by division the formulæ in question. And then, if $u_1+u_2=-(u_3+u_4)$, the functions on the left-hand side become, with only a change of sign, the like functions of u_3+u_4; and we thence have the required equations

$$\frac{s_1-s_4}{c_2d_3-c_3d_2}=-\frac{s_2-s_3}{c_4d_1-c_1d_4},\quad \&c.$$

834.

ON THE ADDITION OF THE ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 56–61.]

MR FORSYTH'S Note [l.c., p. 23] on my "Formula in Elliptic Functions" has supplied a missing link, and I am now able to obtain the addition formulæ very simply from the application of Abel's theorem to the Quadriquadric Curve.

I remark that, instead of coplanar points 1, 2, 3, 4, it is advantageous to consider coresidual points 1, 2 and 3, 4; that is, pairs 1, 2 and 3, 4, which are each of them coplanar with one and the same pair of points 5, 6. The difference is as follows: for the coplanar points 1, 2, 3, 4, we have

$$du_1 + du_2 + du_3 + du_4 = 0,$$

giving

$$u_1 + u_2 + u_3 + u_4 = C,$$

and for the addition theory it is necessary to have $C = 0$; for the coresidual points, we have

$$u_1 + u_2 + u_5 + u_6 = C, \qquad u_3 + u_4 + u_5 + u_6 = C;$$

and thence $u_1 + u_2 = u_3 + u_4$, irrespectively of the value of C.

As to the general theory of a curve in space, observe that, when this is a complete intersection of two surfaces

$$f(x, y, z, w) = 0, \qquad g(x, y, z, w) = 0,$$

then at the point (x, y, z, w), if

$$(x + dx,\ y + dy,\ z + dz,\ w + dw)$$

are the coordinates of the consecutive point, the six coordinates of the tangent line are

$$y\,dz - z\,dy, \quad z\,dx - x\,dz, \quad x\,dy - y\,dx, \quad x\,dw - w\,dx, \quad y\,dw - w\,dy, \quad z\,dw - w\,dz.$$

But considering the line as the intersection of the two tangent planes

$$\frac{df}{dx}X + \frac{df}{dy}Y + \frac{df}{dz}Z + \frac{df}{dw}W = 0,$$

and

$$\frac{dg}{dx}X + \frac{dg}{dy}Y + \frac{dg}{dz}Z + \frac{dg}{dw}W = 0,$$

the six coordinates are

$$\frac{d(f,g)}{d(x,w)},\ \frac{d(f,g)}{d(y,w)},\ \frac{d(f,g)}{d(z,w)},\ \frac{d(f,g)}{d(y,z)},\ \frac{d(f,g)}{d(z,x)},\ \frac{d(f,g)}{d(x,y)},$$

so that the six quotients

$$(y\,dz - z\,dy) \Big/ \frac{d(f,g)}{d(x,w)}, \quad \&\text{c.},$$

are equal to each other, and may be put $= d\omega$.

Considering any two quadric surfaces, there is in general a system of four conjugate points, or points such that in regard to each of the quadrics the polar plane of any one of the points is the plane through the other three points. And then taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ for the equations of the faces of the tetrahedron formed by the four points, the equations of the quadric surfaces will be of the form

$$ax^2 + by^2 + cz^2 + dw^2 = 0,$$

$$a'x^2 + b'y^2 + c'z^2 + d'w^2 = 0;$$

we then have the six quotients

$$(y\,dz - z\,dy) \,/\, (ad' - a'd)\,xw, \quad \&\text{c.},$$

equal to each other, and each $= d\omega$. Here $d\omega$ is homogeneous of the degree zero in the coordinates (x, y, z, w), or, what is the same thing, it is a differential

$$F\Big(\frac{x}{w}, \frac{y}{w}, \frac{z}{w}\Big) d\Big(\frac{x}{w}\Big),$$

say it is $= du$; and taking the integrals always from one and the same fixed point on the curve, we have each point of the curve corresponding to a determinate value of a parameter u.

Supposing that u_1, u_2, u_5, u_6 are the values of u, belonging to any four coplanar points 1, 2, 5, 6; then, by Abel's theorem, $du_1 + du_2 + du_5 + du_6 = 0$; that is, we have

$$u_1 + u_2 + u_5 + u_6 = C,$$

as the condition in order that the four points 1, 2, 5, 6 may be coplanar; similarly, we have

$$u_3 + u_4 + u_5 + u_6 = C,$$

as the condition in order that the four points 3, 4, 5, 6 may be coplanar; and we have therefore

$$u_1 + u_2 = u_3 + u_4,$$

as the condition that the two pairs of points 1, 2 and 3, 4 may be coresidual.

The points 1, 2, 5, 6 are coplanar, hence the line 56 meets the line 12, say in the point A; and the points 3, 4, 5, 6 are coplanar, hence the line 56 meets the line 34, say in the point B. We can, through the curve and any arbitrary point in space, draw a quadric surface

$$(a+\lambda a')\,x^2+(b+\lambda b')\,y^2+(c+\lambda c')\,z^2+(d+\lambda d')\,w^2=0.$$

Hence we have such a quadric surface through the point A; and this surface, passing through 5 and 6, will contain the line 56, and therefore also the point B; hence, passing through 3 and 4, it will contain the line 34; viz. we have the lines 12, 34 as generating lines, obviously of the same kind, on the last-mentioned quadric surface. I say that if, on such a surface, that is, on any surface

$$Ax^2+By^2+Cz^2+Dw^2=0,$$

we have

$$(a,b,c,f,g,h),\quad (a',b',c',f',g',h'),$$

the coordinates of two generating lines of the same kind, then

$$\frac{a}{f}=\frac{a'}{f'},\qquad \frac{b}{g}=\frac{b'}{g'},\qquad \frac{c}{h}=\frac{c'}{h'}.$$

This is at once seen to be the case; for, taking θ an arbitrary parameter, we have for the equations of a generating line

$$\{x\sqrt{A}+iy\sqrt{B}\}+\theta\{z\sqrt{C}+iw\sqrt{D}\}=0,$$
$$\{x\sqrt{A}-iy\sqrt{B}\}-\theta\{z\sqrt{C}-iw\sqrt{D}\}=0,$$

and the coordinates (a,b,c,f,g,h) of this line are

$$-i\sqrt{AD}\,(1-\theta^2),\quad \sqrt{BD}\,(-1-\theta^2),\quad i\sqrt{CD}\cdot 2\theta,\quad \ldots$$

that is, the quotients $\frac{a}{f}$, $\frac{b}{g}$, $\frac{c}{h}$ are each of them independent of θ; and they have consequently their values unaltered when for the original line we substitute any other generating line of the same kind. Or, to prove the statement in a different manner, the equation of the quadric surface through the line (a,b,c,f,g,h) is

$$agh\,x^2+bhf\,y^2+cfg\,z^2+abc\,w^2=0;$$

hence, if this contains the line (a',b',c',f',g',h'), we must have

$$agh:bhf:cfg:abc=a'g'h':b'h'f':c'f'g':a'b'c',$$

equations which give either

$$af+a'f=0,\quad bg+b'g=0,\quad ch+c'h=0,$$

or else

$$af-a'f=0,\quad bg-b'g=0,\quad ch-c'h=0.$$

In the former case, the two lines are generating lines of different kinds; in the latter, they are generating lines of the same kind.

Now, considering (a, b, c, f, g, h) as the coordinates of the line 12 and (a', b', c', f', g', h') as those of the line 34, the equations just obtained are

$$\frac{y_1z_2 - y_2z_1}{x_1w_2 - x_2w_1} = \frac{y_3z_4 - y_4z_3}{x_3w_4 - x_4w_3}, \quad \&c.$$

Of course the equations hold good if, instead of the two lines, we have one and the same line; the equations

$$ax^2 + by^2 + cz^2 + dw^2 = 0, \qquad a'x^2 + b'y^2 + c'z^2 + d'w^2 = 0,$$

considering therein x^2, y^2, z^2, w^2 as coordinates, may be regarded as the equations of a line; and thus the points $(x_1^2, y_1^2, z_1^2, w_1^2)$, &c., will be four points on a line. And we have thus

$$\frac{y_1^2 - y_2^2}{x_1^2 - x_2^2} = \frac{y_3^2 - y_4^2}{x_3^2 - x_4^2}, \quad \&c.,$$

equations which are, by means of the foregoing set, converted into

$$\frac{y_1z_2 - y_2z_1}{x_1w_2 - x_2w_1} = \frac{y_3z_4 - y_4z_3}{x_3w_4 - x_4w_3}, \quad \&c.$$

If for x, y, z, w we write s, c, d, 1, then the equations are

$$\frac{c_1 - c_2}{s_1 - s_2} = \frac{c_3 - c_4}{s_3 - s_4}, \qquad \frac{d_1 - d_2}{s_1 - s_2} = \frac{d_3 - d_4}{s_3 - s_4},$$

where s_1, c_1, d_1 are the sn, cn, and dn of u_1, &c.; and where the relation between the arguments is $u_1 + u_2 = u_3 + u_4$.

In particular, if $u_4 = 0$, we have $s_4 = 0$, $c_4 = 1$, $d_4 = 1$; and then writing u for u_3, and consequently s, c, d for s_3, c_3, d_3, the relation between the arguments is $u = u_1 + u_2$; and we have

$$\frac{c_1 - c_2}{s_1 - s_2} = \frac{c - 1}{s}, \qquad \frac{d_1 - d_2}{s_1 - s_2} = \frac{d - 1}{s}.$$

Combining with these the further relations

$$\frac{c_1d_2 - c_2d_1}{s_1 - s_2} = \frac{cd - 1}{s}, \qquad \frac{d_1s_2 - d_2s_1}{c_1 - c_2} = \frac{d - 1}{c + 1}, \qquad \frac{s_1c_2 + s_2c_1}{d_1 + d_2} = \frac{s}{d + 1},$$

the last two pairs give

$$\frac{1 + c}{1 - c} = \frac{(d_1s_2 - d_2s_1)(d_1 + c_2)}{(d_1s_2 - d_2s_1)(c_1 - c_2)}, \qquad \frac{d + 1}{d - 1} = \frac{(s_1c_2 - s_2c_1)(d_1 + c_2)}{(s_1c_2 - s_2c_1)(c_1 - c_2)};$$

that is,

$$\ldots = \frac{s_1c_2d_2 - s_2c_1d_1}{s_1c_2d_1 - s_2c_1d_2}, \qquad \ldots = \frac{s_1c_2d_2 - s_2c_1d_1}{s_1c_1d_1 - s_2c_2d_2}.$$

[Several intermediate forms of these expressions appear in the scan; the indicated denominators carry the same factor.]

These last values give

$$c - d = \frac{(s_1 + s_2)(c_2 d_1 - c_1 d_2)}{s_1 c_1 d_1 - s_2 c_2 d_2}, \qquad c + d = \frac{(s_1 - s_2)(c_2 d_1 + c_1 d_2)}{s_1 c_1 d_1 - s_2 c_2 d_2};$$

and then, from the given value of either $\dfrac{c-1}{s}$ or $\dfrac{c+1}{s}$, we obtain

$$s = \frac{s_1^2 - s_2^2}{s_1 c_1 d_1 - s_2 c_2 d_2},$$

viz. the resulting equations thus are

$$s = \frac{s_1^2 - s_2^2}{s_1 c_1 d_1 - s_2 c_2 d_2}, \quad c = \frac{s_1 c_1 d_2 - s_2 c_2 d_1}{s_1 c_1 d_1 - s_2 c_2 d_2}, \quad d = \frac{s_1 d_1 c_2 - s_2 d_2 c_1}{s_1 c_1 d_1 - s_2 c_2 d_2},$$

which are one of the four sets given (p. 63) of my *Elliptic Functions*; it may be noticed that they have the advantage of not containing k explicitly, and the disadvantage of becoming vanishing fractions for $u_1 = u_2$. To obtain the ordinary forms, we have only to multiply the numerators and denominators each by $s_1 c_1 d_1 + s_2 c_2 d_2$; the denominator thus becomes $= (s_1^2 - s_2^2)(1 - k^2 s_1^2 s_2^2)$, and each of the numerators has, or acquires, the factor $s_1^2 - s_2^2$, so that this factor divides out.

835.

ON CARDAN'S SOLUTION OF A CUBIC EQUATION.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 96, 97.]

It is interesting to see how the solution comes out when one root of the equation is known. Say the equation is $x^3+qx-r=0$, where $a^3+qa-r=0$, that is, $r=a^3+qa$.

Solving in the usual manner, we have

$$x=y+z, \qquad y^3+z^3-r+(y+z)(3yz+q)=0,$$

whence

$$y^3+z^3=r, \qquad yz=-\tfrac{1}{3}q;$$

and thence

$$(y^3-z^3)^2=r^2+\tfrac{4}{27}q^3=a^6+2qa^4+q^2a^2+\tfrac{4}{27}q^3=(a^2+\tfrac{4}{3}q)(a^2+\tfrac{1}{3}q)^2,$$

or say

$$y^3-z^3=(a^2+\tfrac{1}{3}q)\sqrt{a^2+\tfrac{4}{3}q};$$

and therefore

$$8y^3=4a^3+4qa+2(a^2+\tfrac{1}{3}q)\sqrt{a^2+\tfrac{4}{3}q}=\left\{a+\sqrt{a^2+\tfrac{4}{3}q}\right\}^3,$$

$$8z^3=4a^3+4qa-2(a^2+\tfrac{1}{3}q)\sqrt{a^2+\tfrac{4}{3}q}=\left\{a-\sqrt{a^2+\tfrac{4}{3}q}\right\}^3,$$

where observe that the essential step is the expression of the irrational functions as perfect cubes: that the functions are the cubes of $a\pm\sqrt{a^2+\tfrac{4}{3}q}$ respectively is seen to be true; but if we were to attempt to find a cube root $\alpha+\beta\sqrt{a^2+\tfrac{4}{3}q}$ by an algebraical process, we should be thrown back upon the original cubic equation.

Writing then ω for an imaginary cube root of unity, we have

$$2y=(1,\ \omega,\ \omega^2)\left\{a+\sqrt{a^2+\tfrac{4}{3}q}\right\},$$

$$2z=(1,\ \omega^2,\ \omega)\left\{a-\sqrt{a^2+\tfrac{4}{3}q}\right\};$$

and then

$$x=y+z=a, \quad \text{or} \quad =-\tfrac{1}{2}a\pm\tfrac{1}{2}\sqrt{-3(a^2+\tfrac{4}{3}q)},$$

where $\omega-\omega^2=i\sqrt{3}$; the last two roots are of course the roots of the quadric equation $x^2+ax+a^2+q=0$, which is obtained by throwing out the factor $x-a$ from the given equation $x^3+qx-r=0$.

Cambridge, Sep. 17, 1884.

836.

ON THE QUATERNION EQUATION $qQ - Qq' = 0$.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 108–112.]

I CONSIDER the equation $qQ - Qq' = 0$, where q, q' are given quaternions, and Q is a quaternion to be determined. Obviously a condition must be satisfied by the given quaternions; for, substituting in the given equation for q, q', Q their values, say $w+ix+jy+kz$, $w'+ix'+jy'+kz'$, and $W+iX+jY+kZ$ respectively, and equating to zero the scalar part and the coefficients of i, j, k, we have four equations linear in W, X, Y, Z, and then eliminating these quantities, we have the condition in question. Supposing the condition satisfied, the ratios of W, X, Y, Z are then completely determined, and the required quaternion Q is thus determinate except as to a scalar factor, or say Q is = product of an arbitrary scalar into a determinate quaternion expression.

It might, at first sight, appear that the condition is that the given quaternions shall have their tensors equal, $Tq = Tq'$; for the equation gives $Tq \cdot TQ - TQ \cdot Tq' = 0$, that is, $TQ(Tq - Tq') = 0$. But we cannot thence infer, and it is not true, that the condition is $Tq - Tq' = 0$; the formula does not give the required condition at all, but the conclusion to which it leads is that, when the condition is satisfied, then in general (that is, unless $Tq - Tq' = 0$) the required quaternion is an imaginary quaternion (or, as Hamilton calls it, a biquaternion) having its tensor $TQ = 0$. In the particular case where the given quaternions are such that $Tq - Tq' = 0$, then the required quaternion Q is determined less definitely, viz. it becomes = product of an arbitrary scalar into a not completely determined quaternion expression; and it is thus in general such that TQ is not = 0. In explanation, observe that, for the particular case in question, the four linear equations for W, X, Y, Z reduce themselves to two independent relations, and they give therefore for the ratios of W, X, Y, Z expressions involving an arbitrary parameter λ; these expressions cannot, it is clear, be deduced from the determinate expressions which belong to the general case.

Instead of directly working out the condition in the manner indicated above, I present the investigation in a synthetic form as follows:

Taking v, v' for the vector parts of the two given quaternions, so that $q = w + v$, $q' = w' + v'$, I write for shortness

$$\begin{aligned}
\theta &= w - w', \\
\alpha &= v^2 + v'^2, \quad = -x^2 - y^2 - z^2 - x'^2 - y'^2 - z'^2, \\
\beta &= v^2 - v'^2, \quad = -x^2 - y^2 - z^2 + x'^2 + y'^2 + z'^2, \\
D &= -\theta(\alpha - \theta^2), \\
A &= \beta - \theta^2, \\
B &= \beta + \theta^2,
\end{aligned}$$

so that θ, α, β, D, A, B are all of them scalars. With these I form a quaternion $Q = (D + Av)(D + Bv')$; I say that we have identically

$$qQ - Qq' = \{D - v \cdot v'^2 + v' \cdot v^2 + w \cdot \theta\}\big(\theta^4 - 2\alpha\theta^2 + \beta^2\big).$$

It of course follows that, if $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, then $qQ - Qq' = 0$, viz. $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$ is the condition $\Omega = 0$, for the existence of the required quaternion; and this condition being satisfied, then (omitting the arbitrary scalar factor) the value of the quaternion is $Q = (D + Av)(D + Bv')$, a value giving $T(Q) = 0$, that is, $W^2 + X^2 + Y^2 + Z^2 = 0$. If $\theta = 0$ (that is, $w = w'$), then the condition becomes $\beta = 0$, that is, $x^2 + y^2 + z^2 - x'^2 - y'^2 - z'^2 = 0$; and these two conditions being satisfied, Q ceases to have the determinate value given by the foregoing formula: it has a value involving an arbitrary parameter, and is no longer such that $W^2 + X^2 + Y^2 + Z^2 = 0$.

The identical equation is at once verified; we have

$$\begin{aligned}
qQ - Qq' &= (w + v)Q - Q(w' + v') \\
&= \theta\,(D^2 + DAv + DBv' + ABvv') \\
&\quad + v(D^2 + DAv + DBv' + ABvv') \\
&\quad - (D^2 + DAv + DBv' + ABvv')\,v' \\
&= \theta D^2 + DAv^2 - DBv'^2 \\
&\quad + v(DA\theta + D^2 - ABv'^2) \\
&\quad + v'(DB\theta + ABv^2 - D^2) \\
&\quad + vv'(\theta AB + DB - DA).
\end{aligned}$$

The first line is here $= D\,\{D\theta + Av^2 + Bv'^2\}$, viz. the term in { } is

$$-(\alpha - \theta^2)\theta + (\beta - \theta^2)v^2 - (\beta + \theta^2)v'^2, \quad = -\alpha\theta^2 = \theta \cdot \big(\theta^2 - 2\alpha\big)\,;$$

and similarly each of the other lines contains the factor $\theta^4 - 2\alpha\theta^2 + \beta^2$, and the equation is thus seen to hold good.

The tensor of Q is $= (D^2 - A^2v^2)(D^2 - B^2v'^2)$; and we have

$$\begin{aligned} D^2 - A^2v^2 &= \theta^2(\alpha - \theta^2)^2 - (\beta - \theta^2)^2\, v^2 \\ &= \theta^6 - (2\alpha + v^2)\theta^4 + (\alpha^2 + 2\beta v^2)\theta^2 - \beta^2 v^2, \end{aligned}$$

which, observing that $\alpha^2 + 2\beta v^2 = \beta^2 + 2v^2(2v^2 - \alpha)$ is $= (\theta^2 - v'^2)^2$ and similarly $D^2 - B^2v'^2$ is $= (\theta^2 - v^2)^2$; hence the tensor is

$$TQ = (\theta^2 - v^2)(\theta^2 - v'^2),$$

which, in virtue of $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, is $= 0$.

The particular case is when $Tq - Tq' = 0$, that is, $w^2 - v^2 - w'^2 + v'^2 = 0$, or say $w^2 - w'^2 = v^2 - v'^2$, that is, $\theta(w + w') = \beta$. Combining with this the general condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, we find $\theta^2\{\theta^2 - 2\alpha + (w + w')^2\} = 0$, or in the second factor, for θ^2 and α substituting their values, we have $2\theta^2(w^2 - v^2 + w'^2 - v'^2) = 0$, that is, $2\theta^2(Tq + Tq') = 0$. Attending to the assumed relation $Tq - Tq' = 0$, the second factor can only vanish if $Tq = 0$, $Tq' = 0$; hence, disregarding this more special case, the factor which vanishes must be the first factor, that is, $\theta = 0$; or the equations $Tq - Tq' = 0$ and $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$ give $\beta = 0$ and $\theta = 0$, that is, we have as already mentioned $w - w' = 0$, and $x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$, viz. the given quaternions have their scalars equal, and the squares of their vectors also equal. The equation here is $vQ - Qv' = 0$; and writing $v^2 = v'^2 = -\rho^2$, we see at once that a solution is

$$Q = (-\rho^2 + vv') + \lambda(v + v'),$$

where λ is an arbitrary scalar; in fact, with this value of Q, we have at once vQ and Qv' each

$$= \rho^2(v + v') + \lambda(\rho^2 + vv');$$

and the equation $vQ - Qv' = 0$ is thus satisfied. The value of the tensor is easily found to be

$$TQ = 2(\lambda^2 + \rho^2)(\rho^2 + xx' + yy' + zz'),$$

which is not $= 0$.

In accordance with a remark in the introductory paragraphs, the solution

$$Q = -\rho^2 + vv' + \lambda(v + v')$$

is not comprised in the general solution. As to this, observe that, in the case in question $\theta = 0$, $\beta = 0$, we have from the general theorem the form

$$Q = \left(-\frac{\alpha\theta^2}{B} + v\right)\left(-\frac{\alpha\theta^2}{B} + v'\right),$$

that is,

$$Q = \frac{\alpha^2\theta^4}{B^2} + vv' - \frac{\alpha\theta^2}{B}(v + v').$$

In the condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, writing $\theta = 0$, we have $\theta^2 = 2\alpha$, or for α writing its value $-2\rho^2$, we have $\theta^2 = -4\rho^2$, $\dfrac{\theta^2}{4} = -\rho^2$, and thence $\dfrac{\alpha\theta^2}{B} = -\rho^2$, and $\dfrac{\alpha\theta^2}{B} = \lambda\rho$, if λ denote the $\sqrt{-1}$ of ordinary algebra. The resulting formula is thus

$$Q = -\rho^2 + vv' \pm \lambda\rho(v + v'),$$

which corresponds to the determinate value $\pm\lambda\rho$ of the constant λ.

The foregoing solution $Q = -\rho^2 + vv' + \lambda(v + v')$ may be easily identified with that given pp. 124, 125 of Tait's *Elementary Treatise on Quaternions*; the case there considered is that of a real quaternion, and it was therefore assumed that the two conditions $w - w' = 0$ and $x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$ were each of them satisfied.

The theory of quaternions is, as is well known, identical with that of matrices of the second order; the identity is, in effect, established by the remark and footnote "Linear Associative Algebra," *American Journal of Mathematics*, t. IV. (1881), p. 132. Writing x, y, z, w for Peirce's imaginaries i, j, k, l, these have the multiplication table

	x	y	z	w
x	x	y	0	0
y	0	0	x	y
z	z	w	0	0
w	0	0	z	w

Then if $\lambda_1 = \sqrt{-1}$, be the imaginary of ordinary algebra, and i, j, k the quaternion imaginaries, the relations between i, j, k and x, y, z, w are

$$\begin{aligned} x &= \tfrac{1}{2}(1 + \lambda_1 i), \quad \text{or conversely} \quad 1 = x + w, \\ y &= \tfrac{1}{2}(\ j - \lambda_1 k), \qquad\qquad\qquad i = \lambda_1(x - w), \\ z &= \tfrac{1}{2}(-j - \lambda_1 k), \qquad\qquad\qquad j = (y - z), \\ w &= \tfrac{1}{2}(1 + \lambda_1 i), \qquad\qquad\qquad k = \lambda_1(y + z); \end{aligned}$$

and we can thus at once express a quaternion as a linear function of the x, y, z, w, or a linear function of the x, y, z, w as a quaternion. And we then consider $ax + by + cz + dw$ as denoting the matrix

$$\begin{pmatrix} a, & b \\ c, & d \end{pmatrix},$$

we obtain for the product of two matrices the ordinary formula

$$\begin{pmatrix} a, & b \\ c, & d \end{pmatrix}\begin{pmatrix} a', & b' \\ c', & d' \end{pmatrix} = \begin{pmatrix} (a,b)\cdot(a',c'), & (a,b)\cdot(b',d') \\ (c,d)\cdot(a',c'), & (c,d)\cdot(b',d') \end{pmatrix};$$

viz. we have

$$\begin{aligned}(ax+by+cz+dw)(a'x+b'y+c'z+d'w) &= aa'x^2+bc'yz+\&\text{c.},\\ &= (aa'+bc')\,x+(ab'+bd')\,y\\ &\quad+(ca'+dc')\,z+(cb'+dd')\,w,\end{aligned}$$

in accordance with the formula for the product of the two matrices. Observe that, writing

$$A+Bi+Cj+Dk = A(x+w)+B\lambda_1(x-w)+C(y-z)+D\lambda_1(y+z) = ax+by+cz+dw,$$

we have

$$A=a+d,\quad B\lambda_1=a-d,\quad C=b-c,\quad D\lambda_1=b+c,$$

and thence

$$ad-bc = A^2+B^2+C^2+D^2,$$

so that the determinant of the matrix corresponds to the tensor of the quaternion.

837.

ON THE SO-CALLED D'ALEMBERT–CARNOT GEOMETRICAL PARADOX.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 113, 114.]

THE present note has reference to Prof. Sylvester's paper on this subject [l.c., pp. 92–96]. I cannot admit that D'Alembert and Carnot raised a well-founded objection to "the then and even now too prevalent interpretation of the meaning of the geometrical positive and negative"; it appears to me that the objection was not a well-founded one.

Consider through the origin K an indefinite line $t'Kt$, and measure off from K in the sense Kt a distance equal to the positive quantity a, and let m be the extremity of the distance thus measured off. There is not in the ordinary theory any reason why the distance Km should be $= +a$ rather than $= -a$: it is $= +a$, if Kt be the positive sense of the line through K, and it is $= -a$ if Kt' be the positive sense of the line through K; if it be undetermined which of the two is the positive sense, then the distance Km is $= \pm a$, the sign being essentially indeterminate.

The problem is from a point K outside a given circle to draw a line Kmm' such that the intercepted portion mm' within the circle has a given value c.

Supposing that the line from K to the centre meets the circle in the points A, B at the distances $KA = a$, $KB = b$: then if $Km = r$, we have $r(c - r) = ab$, or $r = \frac{1}{2}c \pm \sqrt{\frac{1}{4}c^2 + ab}$; viz. we have for r, not simultaneously but alternatively, the positive value $\frac{1}{2}c + \sqrt{\frac{1}{4}c^2 + ab}$, and the negative value $\frac{1}{2}c - \sqrt{\frac{1}{4}c^2 + ab}$, the latter of these being the greatest in absolute magnitude; say the values are $+\rho_1$ and $-\rho_2$. We may with either of these values construct the point m; viz. we obtain m as one of the intersections of the given circle with the circle centre K and radius ρ_1, or else with the circle centre K and radius ρ_2 (that is, radius $-\rho_2$); and attending to the intersections on the same side of the line from K to the centre, it happens that

[Article 837 continues on the following pages, which lie outside the present chunk.]